

```
PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker image ls
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
docker-gs-ping       multistage   a0563d1eed2    55 seconds ago 48.2MB
docker/welcome-to-docker latest       eedaff45e3c7   20 months ago 29.5MB

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping>

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker run docker-gs-ping

  ____  _
 / ___|| | | |
| |___| |_| |
 \___|_____|_|_|
v4.10.2
High performance, minimalist Go web framework
https://echo.labstack.com

-----o/-----
      O\

  http server started on [::]:8080

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2> curl http://localhost:8080/
<!doctype html><html lang="en"><head><meta charset="utf-8"><link rel="icon" href="/favicon.ico"/><meta name="viewport" content="width=device-width,initial-scale=1"/><meta
name="theme-color" content="#000000"/><meta name="description" content="Your first Docker container"/><link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesomsome/6.4.2/css/all.min.css"/><title>Welcome to Docker</title><script defer="defer" src="/static/js/main.c9e951e4.js"></script><link href="/static/css/main.27312bf9.cs
s" rel="stylesheet"></head><body><noscript>You need to enable JavaScript to run this app.</noscript><div id="root"></div></body></html>

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2>
```

```
PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
36faf73e4839   docker-gs-ping  "/docker-gs-ping"        2 minutes ago  Up 2 minutes  8080/tcp                           tender_bhabha
4cc5bd0038da   docker/welcome-to-docker  "/docker-entrypoint..." 26 hours ago  Up 26 hours   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  jovial_hopper

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping>
```

```
PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker stop tender_bhabha
tender_bhabha

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping>
```

```
PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker run -d -p 8080:8080 --name rest-server docker-gs-ping
0c28e8c011ebace76ff91dd4d63d2edc2bb2259d167c052c5a8829c8709c1869

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
0c28e8c011eb   docker-gs-ping  "/docker-gs-ping"        14 seconds ago  Up 13 seconds  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp  rest-server

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping>
```

```
PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS pwsh - d

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker image tag docker-gs-ping:latest docker-gs-ping:v1.0
PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker images
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
docker-gs-ping       latest       db0f5c8c826b   2 minutes ago  1.69GB
docker-gs-ping       v1.0        db0f5c8c826b   2 minutes ago  1.69GB
docker/welcome-to-docker latest       eedaff45e3c7   20 months ago 29.5MB

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping>
```

```

● PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker image rm docker-gs-ping
Untagged: docker-gs-ping:latest
Deleted: sha256:db0f5c8c826b73bd68bfd3e557c7c94cff6f6c7a7ea231210825972ea9965159
● PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
docker/welcome-to-docker latest             eedaff45e3c7       20 months ago     29.5MB
○ PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> █

```

```

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker run -d -p 8080:8080 docker-gs-ping
689e37fb2005c9c69f51101ced5ec92d0107e8a450a3caf22ab3efc21022740
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint quirky_williams (44d8e2d6aceedf87f0b2
065ed432d263158de8f37656508f5f0fc0583925867e): Bind for 0.0.0.0:8080 failed: port is already allocated

Run 'docker run --help' for more information
● PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker run -d 8080:8080 docker-gs-ping
Unable to find image '8080:8080' locally
docker: Error response from daemon: pull access denied for 8080, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
● PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker run -d docker-gs-ping
36faf73e48391b2f35e620e915fb677d6b68a5a0e932c34e3876709563a99f4
● PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> curl http://localhost:8080/
<!doctype html><html lang="en"><head><meta charset="utf-8"><link rel="icon" href="/favicon.ico"/><meta name="viewport" content="width=device-width,initial-scale=1"/><meta
name="theme-color" content="#000000"/><meta name="description" content="Your first Docker container"/><link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/fo
nt-awesome/6.4.2/css/all.min.css"/><title>Welcome to Docker</title><script defer="defer" src="/static/js/main.c9e951e4.js"></script><link href="/static/css/main.27312bf9.cs
s" rel="stylesheet"></head><body><noscript>You need to enable JavaScript to run this app.</noscript><div id="root"></div></body></html>
○ PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> █

```

```

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS                               NAMES
36faf73e4839   docker-gs-ping      "/docker-gs-ping"       2 minutes ago   Up 2 minutes   8080/tcp                           tender_bhabha
4cc5bd0038da   docker/welcome-to-docker "/docker-entrypoint..." 26 hours ago   Up 26 hours   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp jovial_hopper
● PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker stop tender_bhabha
Error response from daemon: No such container: tender_bhabha
● PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker stop tender_bhabha
tender_bhabha
○ PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> █

```

```

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS                               NAMES
4cc5bd0038da   docker/welcome-to-docker "/docker-entrypoint..." 26 hours ago   Up 26 hours   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp jovial_hopper
● PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker ps --all
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS                               NAMES
36faf73e4839   docker-gs-ping      "/docker-gs-ping"       5 minutes ago   Exited (2) About a minute ago                                tender_bhabha
689e37fb2005   docker-gs-ping      "/docker-gs-ping"       5 minutes ago   Created                                           quirky_williams
26fe704fd1a5   docker-gs-ping      "/docker-gs-ping"       6 minutes ago   Created                                           laughing_merkle
833fe5e7d45b   docker-gs-ping      "/docker-gs-ping"       9 minutes ago   Exited (2) 6 minutes ago                                vibrant_wilbur
4cc5bd0038da   docker/welcome-to-docker "/docker-entrypoint..." 26 hours ago   Up 26 hours   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp jovial_hopper
○ PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> █

```

```

● PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker restart tender_bhabha
tender_bhabha
● PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker ps --all
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS                               NAMES
36faf73e4839   docker-gs-ping      "/docker-gs-ping"       6 minutes ago   Up 6 seconds   8080/tcp                           tender_bhabha
689e37fb2005   docker-gs-ping      "/docker-gs-ping"       7 minutes ago   Created                                           quirky_williams
26fe704fd1a5   docker-gs-ping      "/docker-gs-ping"       8 minutes ago   Created                                           laughing_merkle
833fe5e7d45b   docker-gs-ping      "/docker-gs-ping"       11 minutes ago   Exited (2) 8 minutes ago                                vibrant_wilbur
4cc5bd0038da   docker/welcome-to-docker "/docker-entrypoint..." 26 hours ago   Up 26 hours   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp jovial_hopper
○ PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> █

```

```

PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker rm tender_bhabha quirky_williams laughing_merkle vibrant_wilbur jovial_hopper
tender_bhabha
quirky_williams
laughing_merkle
vibrant_wilbur
jovial_hopper
PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker ps --all
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS                               NAMES
PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> █

```

```

Welcome  main.go 5 X Dockerfile 1 Extension: Container Tools
main.go > ...
1 package main
2
3 import (
4     "net/http"
5     "os"
6
7     "github.com/labstack/echo/v4"
8     "github.com/labstack/echo/v4/middleware"
9 )
10
11 func main() {
12
13     e := echo.New()
14
15     e.Use(middleware.Logger())
16     e.Use(middleware.Recover())
17
18     e.GET("/", func(c echo.Context) error {
19         return c.HTML(http.StatusOK, "Hello, Docker! <3")
20     })
21
22     e.GET("/health", func(c echo.Context) error {
23         return c.JSON(http.StatusOK, struct{ Status string }(Status: "OK"))
24     })
25
26     httpPort := os.Getenv("PORT")
27     if httpPort == "" {
28         httpPort = "8080"
29     }
30
31     e.Logger.Fatal(e.Start(":" + httpPort))
32 }
33
34 // Simple implementation of an integer minimum
35 // Adapted from: https://gobyexample.com/testing-and-benchmarking
36 func IntMin(a, b int) int {
37     if a < b {
38         return a
39     }
40     return b
41 }
42
```

Dockerfile > ...

```
1 # syntax=docker/dockerfile:1
2
3 FROM golang:1.19
4
5 # Set destination for COPY
6 WORKDIR /app
7
8 # Download Go modules
9 COPY go.mod go.sum ./
10 RUN go mod download
11
12 # Copy the source code. Note the slash at the end, as explained in
13 # https://docs.docker.com/reference/dockerfile/#copy
14 COPY *.go ./
15
16 # Build
17 RUN CGO_ENABLED=0 GOOS=linux go build -o /docker-gs-ping
18
19 # Optional:
20 # To bind to a TCP port, runtime parameters must be supplied to the docker command.
21 # But we can document in the Dockerfile what ports
22 # the application is going to listen on by default.
23 # https://docs.docker.com/reference/dockerfile/#expose
24 EXPOSE 8080
25
26 # Run
27 CMD ["/docker-gs-ping"]
```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
[+] Building 16.5s (6/12)
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 727B
=> resolve image config for docker-image://docker.io/docker/dockerfile:1
=> docker-image://docker.io/docker/dockerfile:1@sha256:9857836c9ee4268391bb5b09f9f157f3c91bb15821bb77969642813b0d00518d
=> => resolve docker.io/docker/dockerfile:1@sha256:9857836c9ee4268391bb5b09f9f157f3c91bb15821bb77969642813b0d00518d
=> => sha256:724f0f61ffc85cf6ad0032c9063f7a74cd6d3590fd1138154820556e0a92073b 13.16MB / 13.16MB
=> => extracting sha256:724f0f61ffc85cf6ad0032c9063f7a74cd6d3590fd1138154820556e0a92073b
=> [internal] load metadata for docker.io/library/golang:1.19
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/6] FROM docker.io/library/golang:1.19@sha256:3025bf670b8363ec9f1b4c4f27348e6d9b7fec607c47e401e40df816853e743a
=> => resolve docker.io/library/golang:1.19@sha256:3025bf670b8363ec9f1b4c4f27348e6d9b7fec607c47e401e40df816853e743a
=> => sha256:5ec11cb68eac452710eadb46df5e6cf6ead699755303758bf1e262e47b013417 155B / 155B
=> => sha256:0008c64687902329ac64331848304f6f0bc14b86d6be46bccf79292b564f6587 15.73MB / 149.14MB
=> => sha256:190fa1651026077cee00b53a754acbe0dc734b99255c6b274e798f6cd877ae18 15.73MB / 92.27MB
=> => sha256:9f13f5a53d118643c1f1ff294867c09f224d00edca21f56caa71c2321f8ca004 17.83MB / 64.11MB
=> => sha256:00046d1e755ea94fa55a700ca9a10597e4fac7c47be19d970a359b0267a51fbf 24.03MB / 24.03MB
=> => sha256:012c0b3e998c1a0c0bedcf712eaaafb188580529dd026a04aa1ce13fdb39e42b 12.58MB / 49.56MB
=> [internal] load build context
=> => transferring context: 6.22kB
```

```

=> => transferring context: 6.22kB
=> [2/6] WORKDIR /app
=> [3/6] COPY go.mod go.sum ./
=> [4/6] RUN go mod download
=> [5/6] COPY *.go ./
=> [6/6] RUN CGO_ENABLED=0 GOOS=linux go build -o /docker-gs-ping
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:646cac43dcd445f5dddc06efa6d1c175cfa00cba168ce9699fcd2433222cfaa6
=> => exporting config sha256:5ae4d76156782ab808dcd536d3ea937619ebd9942e8de21c6cc19543fb18c30c
=> => exporting attestation manifest sha256:6c691728110d094e0869344125034cee250791413d293d73c7da18d09d41b59
=> => exporting manifest list sha256:db0f5c8c826b73bd68bfd3e557c7c94cff6f6c7a7ea231210825972ea9965159
=> => naming to docker.io/library/docker-gs-ping:latest
=> => unpacking to docker.io/library/docker-gs-ping:latest
● PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker image ls
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
docker-gs-ping       latest       db0f5c8c826b   About a minute ago   1.69GB
docker/welcome-to-docker latest       eedaff45e3c7   20 months ago      29.5MB
● PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping> docker images
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
docker-gs-ping       latest       db0f5c8c826b   About a minute ago   1.69GB
docker/welcome-to-docker latest       eedaff45e3c7   20 months ago      29.5MB
○ PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping>

=> => extracting sha256:5664b15f108bf9436ce3312090a767300800edbbfd4511aa1a6d64357024d5dd
=> => extracting sha256:27be814a09ebd97fac6fb7b82d19f117185e90601009df3fbab6f442f85cd6b3
=> => extracting sha256:4aa0ea1413d37a58615488592a0b827ea4b2e48fa5a77cf707d0e35f025e613f
=> => extracting sha256:9ef7d74bdfdf3c517b28bd694a9159e94e5f53ff1ca87b39f8ca1ac0be2ed317
=> => extracting sha256:9112d77ee5b16873acaa186b816c3c61f5f8eba40730e729e9614a27f40211e0
=> => extracting sha256:83f8d4690e1f293d0438aef7d1075e590ce77fdec97bb4d90b1d227aeba343fd
=> => extracting sha256:a4ba90834fb4abf3d80bbdaaaef36560ab1bb682f5279d44114d768e119639b9
=> => extracting sha256:df368711b36276ed02b2040d3e3296b919042d2a05a2bbe9f758e708436c12cf
=> CACHED [build-stage 2/6] WORKDIR /app
=> CACHED [build-stage 3/6] COPY go.mod go.sum ./
=> CACHED [build-stage 4/6] RUN go mod download
=> CACHED [build-stage 5/6] COPY *.go ./
=> CACHED [build-stage 6/6] RUN CGO_ENABLED=0 GOOS=linux go build -o /docker-gs-ping
=> [build-release-stage 2/3] COPY --from=build-stage /docker-gs-ping /docker-gs-ping
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:045942a4c82f77f8d50c173f0c6fd547fcb34ce53b3cd5d1add87005f98bdaf
=> => exporting config sha256:ce2b6349fef91a8ef0d192fa46fe018de7ad7fa929281fab94116310ac5393b
=> => exporting attestation manifest sha256:0524896b0e7f62ca03b50d64b39339b171b885be87706a4a6ff3383aabe809
=> => exporting manifest list sha256:a0563d1eed2b66bf08f059b6b0a0828de233f8eaa8c88fb87324b84caef2e2c
=> => naming to docker.io/library/docker-gs-ping:multistage
=> => unpacking to docker.io/library/docker-gs-ping:multistage
PS C:\Users\RAMIN\Documents\goClass\goClass\week89\lab2\docker-gs-ping>

```